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Critical Analysis of the Energy Landscape in Nunavut (2021)

1. Introduction

The newest province in Canada's history, Nunavut, is the biggest and most northern. On April 1, 1999, it formally became independent from the Northwest Territory. The far north is the coldest part of Canada, covering an enormous 2,038,700 km². With only roughly 36,858 residents as of 2021, it is also the least inhabited province in Canada.

In 2021, Nunavut faced energy challenges due to its terrain, climate, and lack of infrastructure. The region, made up of small, isolated communities, is not connected to national electrical grids or natural gas pipelines. As a result, virtually all energy needs, such as space heating and power, are supplied by diesel fuel, which is shipped in and stored annually in summer.

Nunavut's energy supply consists predominantly of fossil fuels, in the guise of diesel fuel. According to the Canada Energy Regulator (CER), nearly all Nunavut electricity comes from diesel, imported during the summer and stored away for the rest of the year.

Nunavut lacks a natural gas pipeline or electricity links to other governments and has 25 communities with separate, independent diesel power plants supplied by QEC. In 2021, Nunavut produced about 0.28 terawatt-hours of electricity and was below less than 0. Nunavut illustrates where energy security and environmental policy intersect in the remote northern regions.

Nunavut energy development has been constrained traditionally by its physical remoteness and reliance on federal administration. A federal moratorium on offshore oil and gas licensing in the Canadian Arctic, until 2028, constrains local energy development once more. But the 2024 Nunavut Lands and Resources Devolution Agreement is a shift toward territorial control of public land and natural resources. This change in ownership, although incremental, can potentially begin to open the door toward more appropriate policy innovations in the specific circumstances of Nunavut.

In 2021, key themes in energy policy included improving infrastructure resilience, managing high emissions intensity, and addressing long-standing dependence on diesel. Nunavut's 2021 energy policy primarily focused on three key areas. The Government of Nunavut, QEC, Indigenous leaders, and the federal government were included among the stakeholders, such as the renewable energy developers and the mining companies and local communities. Although QEC serves as the primary operator and planner, decisions are increasingly influenced by community-level needs and federal priorities.

The territory's electricity supply is fully dependent on diesel-fired generation. Each of the 25 communities operates its own microgrid, powered by locally stored fuel. In 2021, Nunavut generated only 0.28 terawatt-hours of electricity—less than 0.05% of Canada's total. Total end-use energy demand stood at 3.9 PJ, with nearly 80% derived from refined petroleum products (RPPs).

Transportation and industrial sectors were the largest consumers. Per capita diesel use was the highest in the country, a reflection of both climate demands and inefficient systems. This dependence is illustrated in the energy Sankey diagram, where all energy flows stem from petroleum imports. No other fuel source—renewable or otherwise—appears in the diagram, underscoring the lack of energy diversification.

2. Key Policies and Initiatives

Replacement of aging infrastructure was a priority in Nunavut's energy policy. New diesel plant development in Arctic Bay and system upgrades in Grise Fiord and Cape Dorset were incremental but

vital developments. QEC's multi-year corporate plan outlined similar developments for the remainder of the communities.

Due to a high dependence on fossil fuels, Nunavut had the highest emissions intensity in Canada. The territory was not faced with carbon pricing, but territorial efforts were on adaptation and infrastructure resilience rather than mitigation. In 2022, over half of all the total GHG emissions were from transportation.

Although territorial control over natural resources is increasing, QEC remains the main energy authority. Broader regulatory reforms were modest in 2021, and most policy action was focused on project implementation rather than systemic reform.

Although small-scale renewable initiatives exist—for instance, solar installations in Iqaluit and a hybrid solar-diesel project under development in Kugluktuk—their overall significance in Nunavut's energy blend remains negligible. Five other communities secured federal funding for similar solar projects, but these initiatives are at the initial stages.

Nunavut faces a number of barriers to energy transition: logistical hurdles, limited local technical expertise, and high capital costs. The Sankey diagram reflects this reality. Despite ongoing policy interest in renewables, the actual energy flow in 2021 remained entirely petroleum-based, showing little evidence of transition on the ground.

There is growing local interest in renewables, driven by concern over the environmental and economic expense of diesel. Yet there is skepticism over whether clean alternatives can equal the reliability, particularly during long Arctic winters.

3. Critique of Energy Policy Framework

Nunavut's energy strategy is a model of prudent long-term planning, especially in the context of infrastructure maintenance and territorial energy autonomy. The particular circumstances of the region are also recognized explicitly. Planned outcomes are stingy in fulfillment. Excessive dependence on a sole source of energy poses economic, environmental, and logistical risks for the territory. The gap between policy intent and action is most evident in the glacial pace of integrating renewables.

Energy costs in Nunavut are among the highest in Canada and are symptomatic of a broader issue of accessibility and affordability. The environmental effects, such as the emissions of black carbon, are particularly harmful in Arctic ecosystems. Socially, power disruptions and heating instability are persistent problems in all but a few communities. Sankey diagram of carbon, using significant figures on the MMT scale, also illustrates the energy-emissions relationship. Though emissions are relatively low in value, absolutely speaking, they are disproportionately large per unit of energy consumed.

The COVID-19 pandemic disrupted infrastructure developments and supply chains, delaying timelines for renewable and conventional energy ventures. Struggles highlighted the weakness of Nunavut's supply-based framework. Despite adversity, the territory has shown adaptability with pilot projects and grassroots participation. Hybrids, solar plants, and retiring old diesel engines are encouraging starts.

4. Conclusion and Recommendations

Nunavut's energy system is the product of necessity: inaccessibility, extreme climate, and lack of resource diversity provide scant choice but to depend on diesel. But such dependence is not permanent, nor is it cost-effective.

Policy recommendations

1. Develop region-specific data infrastructure to support planning and accountability.
2. Increase federal funding to reflect the costly nature of Arctic infrastructure.
3. Develop a trained local workforce to drive hybrid and renewable systems.
4. Embed community-led renewable pilots to build trust and produce localized data.

Looking ahead, Nunavut can become a testbed for decentralized, low-emission energy systems suitable for cold and remote regions. With sustained investment and a shift toward Indigenous-led planning, the territory could lead in demonstrating what Arctic resilience looks like in practice.

(1091 words)

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